

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage using HTML Components

2. AIM

To design and develop a webpage using various HTML and CSS components.

To understand the structure and usage of HTML elements for creating a well-organized webpage and styling it using CSS3.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure and components of HTML.
 - To design a well-organized webpage using various HTML elements.
 - To understand and apply CSS3 for styling webpage components.
 - To use HTML elements such as headings, paragraphs, lists, tables, links, images, and forms.
 - To create a visually structured and well-styled webpage using HTML and CSS3.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure webpages. **CSS3 (Cascading Style Sheets)** is used to style and visually enhance the HTML elements of a webpage.

HTML elements/components organize and display different types of content, while CSS3 properties and selectors control their appearance, layout, colors, fonts, spacing, and other visual aspects.

In this application, HTML is used to create and structure the static webpage, while CSS3 is used to style and format the webpage using different components.

- Headings are used to display titles and headings.
- Paragraphs are used to display textual content.
- Lists are used to organize information.
- Images are used to display visual content.
- Links are used to navigate to other webpages.
- Tables are used to display data in rows and columns.
- Forms are used to collect user information.
- CSS3 is used to style HTML elements by applying colors, fonts, borders, spacing, alignment, backgrounds, and layouts.

The basic flow of the webpage is:

HTML Components → Webpage Structure → CSS3 Styling → Browser → Displayed Styled Webpage

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
 2. Add the basic HTML5 document structure using `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>`.
 3. Add metadata and the webpage title inside the `<head>` section.
 4. Create a navigation bar using `<header>`, `<nav>`, `<ul>`, `<li>`, and `<a>` elements.
 5. Create a hero section using `<section>`, headings, paragraphs, and a navigation link.
 6. Create the main webpage content using the `<main>` element.
 7. Create a Features section using `<section>` and `<article>` elements.
 8. Add feature cards containing headings, paragraphs, and icons.
 9. Create an About section using `<section>` and `<div>` elements.
 10. Create a Contact section containing descriptive text and a contact link.
 11. Create a footer using the `<footer>` element.
 12. Apply CSS3 styling using selectors, colors, fonts, backgrounds, borders, spacing, and shadows.
 13. Use CSS Flexbox and Grid to organize the webpage layout.
 14. Apply CSS gradients, transitions, hover effects, and responsive media queries.
 15. Save the HTML file and open it in a web browser.
 16. Verify the webpage structure, navigation links, styling, responsiveness, and overall displayed output.
-

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>TealWave - Modern Webpage</title>

<style>
  /* ----- RESET ----- */
  * {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
  }

  html {
    scroll-behavior: smooth;
  }

  body {
    font-family: Arial, Helvetica, sans-serif;
    background: #f0fbfc;
    color: #12343b;
    line-height: 1.6;
  }

  /* ----- NAVBAR ----- */
  header {
    position: sticky;
    top: 0;
    z-index: 1000;

    background: rgba(255, 255, 255, 0.9);
    backdrop-filter: blur(12px);

    border-bottom: 1px solid #d4eefe;
  }

  nav {
    max-width: 1200px;
    margin: auto;

    display: flex;
    justify-content: space-between;
    align-items: center;

    padding: 18px 30px;
  }

```

```
.logo {
  font-size: 25px;
  font-weight: bold;
  color: #008c95;
}

.logo span {
  color: #006d77;
}

nav ul {
  display: flex;
  list-style: none;
  gap: 30px;
}

nav a {
  text-decoration: none;
  color: #24545b;
  font-weight: 600;
  transition: 0.3s;
}

nav a:hover {
  color: #0097a7;
}

/* ----- HERO ----- */
.hero {
  min-height: 90vh;

  display: flex;
  justify-content: center;
  align-items: center;

  text-align: center;
  padding: 60px 20px;

  background:
    radial-gradient(circle at top right, #5ee7e1 0%, transparent 30%),
    radial-gradient(circle at bottom left, #9be7e5 0%, transparent 30%),
    linear-gradient(135deg, #006d77, #0097a7, #00b4b8);

  color: white;
}

.hero-content {
  max-width: 850px;
}

.hero h1 {
  font-size: clamp(45px, 7vw, 80px);
  margin-bottom: 20px;
  letter-spacing: 2px;
}
```

```
}

.hero h1 span {
  color: #bffffc;
}

.hero p {
  font-size: 20px;
  max-width: 650px;
  margin: 0 auto 35px;
  color: #e4ffff;
}

.btn {
  display: inline-block;

  padding: 14px 30px;
  border-radius: 30px;

  background: white;
  color: #007c83;

  text-decoration: none;
  font-weight: bold;

  transition: 0.3s;
  box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}

.btn:hover {
  transform: translateY(-5px);
  box-shadow: 0 12px 30px rgba(0, 0, 0, 0.25);
}

/* ----- SECTION ----- */
section {
  padding: 80px 30px;
}

.section-title {
  text-align: center;
  margin-bottom: 50px;
}

.section-title h2 {
  font-size: 38px;
  color: #006d77;
  margin-bottom: 10px;
}

.section-title p {
  color: #58777c;
}
```

```
/* ----- CARDS ----- */
.cards {
  max-width: 1100px;
  margin: auto;

  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 25px;
}

.card {
  background: white;
  padding: 35px 28px;

  border-radius: 20px;

  border: 1px solid #d6eeee;

  box-shadow: 0 10px 30px rgba(0, 100, 110, 0.08);

  transition: 0.3s;
}

.card:hover {
  transform: translateY(-10px);
  box-shadow: 0 20px 40px rgba(0, 130, 140, 0.18);
}

.icon {
  width: 60px;
  height: 60px;

  display: flex;
  justify-content: center;
  align-items: center;

  border-radius: 15px;

  background: #dffafa;
  color: #008c95;

  font-size: 28px;

  margin-bottom: 20px;
}

.card h3 {
  color: #006d77;
  margin-bottom: 12px;
  font-size: 22px;
}

.card p {
  color: #607d82;
```

```

}

/* ----- ABOUT ----- */
.about {
    background: #e5f8f8;
}

.about-container {
    max-width: 1000px;
    margin: auto;

    display: grid;
    grid-template-columns: 1fr 1fr;
    gap: 50px;
    align-items: center;
}

.about-text h2 {
    font-size: 38px;
    color: #006d77;
    margin-bottom: 20px;
}

.about-text p {
    color: #526f74;
    margin-bottom: 15px;
}

.about-box {
    padding: 40px;

    border-radius: 25px;

    background: linear-gradient(
        135deg,
        #007c83,
        #00a8ad
    );

    color: white;

    box-shadow: 0 20px 40px rgba(0, 110, 120, 0.2);
}

.about-box h3 {
    font-size: 28px;
    margin-bottom: 15px;
}

/* ----- CONTACT ----- */
.contact-box {
    max-width: 800px;
    margin: auto;

```

```

    text-align: center;

    padding: 50px 30px;

    border-radius: 25px;

    background: white;

    box-shadow: 0 15px 40px rgba(0, 100, 110, 0.12);
}

.contact-box h2 {
    color: #006d77;
    margin-bottom: 15px;
    font-size: 35px;
}

.contact-box p {
    color: #607d82;
    margin-bottom: 25px;
}

.contact-btn {
    display: inline-block;

    padding: 13px 28px;

    border-radius: 25px;

    background: #008c95;
    color: white;

    text-decoration: none;
    font-weight: bold;

    transition: 0.3s;
}

.contact-btn:hover {
    background: #006d77;
    transform: scale(1.05);
}

/* ----- FOOTER ----- */
footer {
    text-align: center;
    padding: 25px;

    background: #004f58;
    color: #d8ffff;
}

/* ----- RESPONSIVE ----- */
@media (max-width: 800px) {

```



```

    nav {
      flex-direction: column;
      gap: 15px;
    }

    nav ul {
      gap: 15px;
    }

    .cards {
      grid-template-columns: 1fr;
    }

    .about-container {
      grid-template-columns: 1fr;
    }

    .hero {
      min-height: 80vh;
    }
  }

  @media (max-width: 500px) {

    nav ul {
      flex-wrap: wrap;
      justify-content: center;
    }

    section {
      padding: 60px 20px;
    }

    .hero p {
      font-size: 17px;
    }
  }
</style>
</head>

<body>

  <!-- NAVIGATION BAR -->
  <header>
    <nav>
      <div class="logo">
        Teal<span>Wave</span>
      </div>

      <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#features">Features</a></li>
        <li><a href="#about">About</a></li>

```

```

        <li><a href="#contact">Contact</a></li>
    </ul>
</nav>
</header>

<!-- HERO SECTION -->
<main>

    <section class="hero" id="home">
        <div class="hero-content">

            <h1>
                Design the <span>Future</span>
            </h1>

            <p>
                A beautiful and modern webpage designed using
                HTML5 and CSS3 with an elegant teal-blue theme.
            </p>

            <a href="#features" class="btn">
                Explore More →
            </a>

        </div>
    </section>

    <!-- FEATURES -->
    <section id="features">

        <div class="section-title">
            <h2>Our Features</h2>
            <p>Simple design. Powerful presentation.</p>
        </div>

        <div class="cards">

            <article class="card">
                <div class="icon">◆</div>

                <h3>Modern Design</h3>

                <p>
                    A clean and attractive interface created
                    using modern HTML5 and CSS3 techniques.
                </p>
            </article>

            <article class="card">
                <div class="icon">⚡</div>

```

```
<h3>Fast & Smooth</h3>
```

```
<p>
```

```
  Lightweight CSS with smooth transitions  
  and hover effects for an interactive experience.
```

```
</p>
```

```
</article>
```

```
<article class="card">
```

```
  <div class="icon">⌘</div>
```

```
  <h3>Responsive</h3>
```

```
  <p>
```

```
    The webpage automatically adapts to  
    desktops, tablets and mobile devices.
```

```
  </p>
```

```
</article>
```

```
</div>
```

```
</section>
```

```
<!-- ABOUT -->
```

```
<section class="about" id="about">
```

```
  <div class="about-container">
```

```
    <div class="about-text">
```

```
      <h2>About This Design</h2>
```

```
      <p>
```

```
        This webpage demonstrates the use of  
        HTML5 semantic elements and CSS3 styling.
```

```
      </p>
```

```
      <p>
```

```
        The teal and blue color palette provides  
        a professional, calm and modern appearance.
```

```
      </p>
```

```
    </div>
```

```
  <div class="about-box">
```

```
    <h3>HTML5 + CSS3</h3>
```

```
    <p>
```

```
      Semantic structure, gradients, shadows,  
      responsive layouts, transitions and
```

```
        modern UI elements are combined to create
        this webpage.
    </p>

</div>

</div>

</section>

<!-- CONTACT -->
<section id="contact">

    <div class="contact-box">

        <h2>Let's Connect</h2>

        <p>
            Have an idea? Let's create something amazing
            together.
        </p>

        <a href="mailto:example@email.com" class="contact-btn">
            Contact Us
        </a>

    </div>

</section>

</main>

<!-- FOOTER -->
<footer>
    <p>
        © 2026 TealWave | Designed using HTML5 & CSS3
    </p>
</footer>

</body>
</html>
```

7. CODE EXPLANATION

HTML/CSS Component	Description
<!DOCTYPE html>	Defines the document as an HTML5 webpage.
<html>	Represents the root element of the HTML document.
<head>	Contains metadata, the webpage title, and CSS styling.
<meta>	Defines character encoding and viewport settings for responsive display.
<title>	Defines the title of the webpage shown in the browser tab.
<style>	Contains the CSS3 rules used to style the webpage.
<body>	Contains all the visible content of the webpage.
<header>	Defines the header section containing the navigation bar.
<nav>	Contains navigation links for different sections of the webpage.
/	Creates and organizes the navigation menu items.
	Creates hyperlinks for navigation between webpage sections and email contact.
<main>	Represents the main content of the webpage.
<section>	Defines separate sections such as Home, Features, About, and Contact.
<article>	Defines independent feature cards containing related content.
<div>	Groups and organizes webpage content for styling and layout.
<h1>, <h2>, <h3>	Define headings and titles at different levels.
<p>	Displays textual and descriptive content.
CSS Selectors	Select HTML elements, classes, and IDs to apply styling.
CSS Box Model	Controls margins, padding, borders, width, and spacing of elements.
Flexbox	Creates flexible layouts for the navigation bar and other components.
CSS Grid	Arranges feature cards and the About section into structured layouts.
CSS Gradients	Creates gradient backgrounds for the hero and About sections.

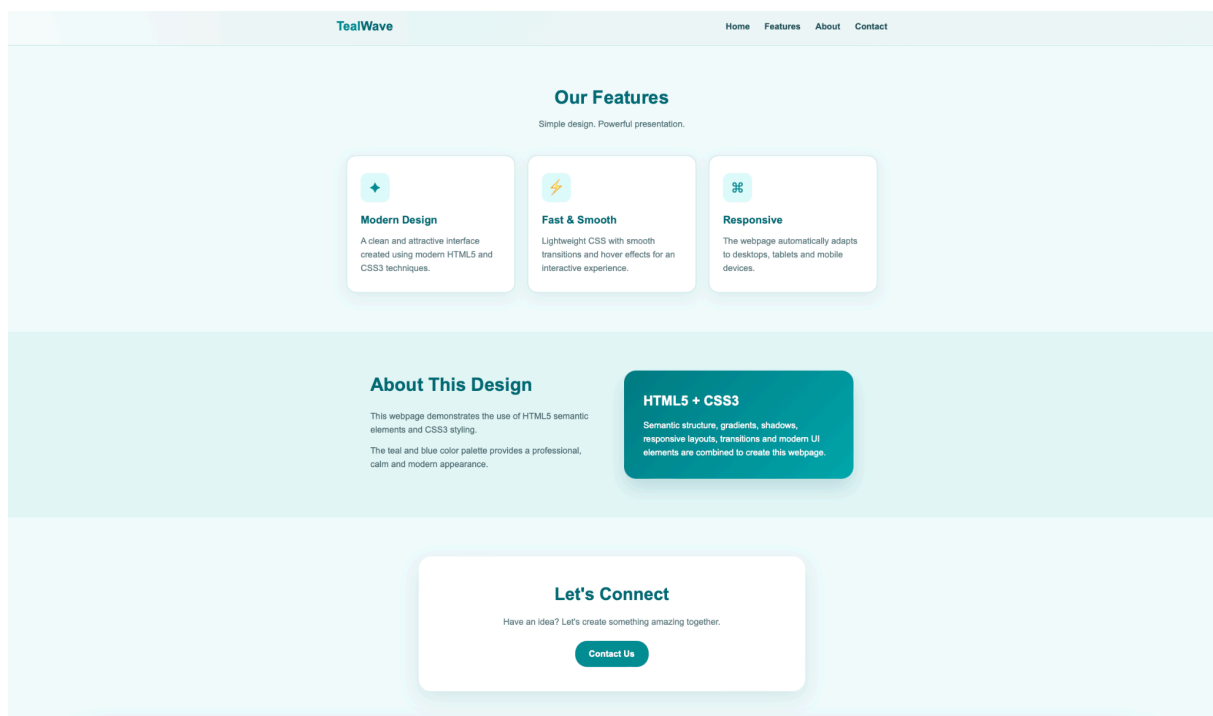
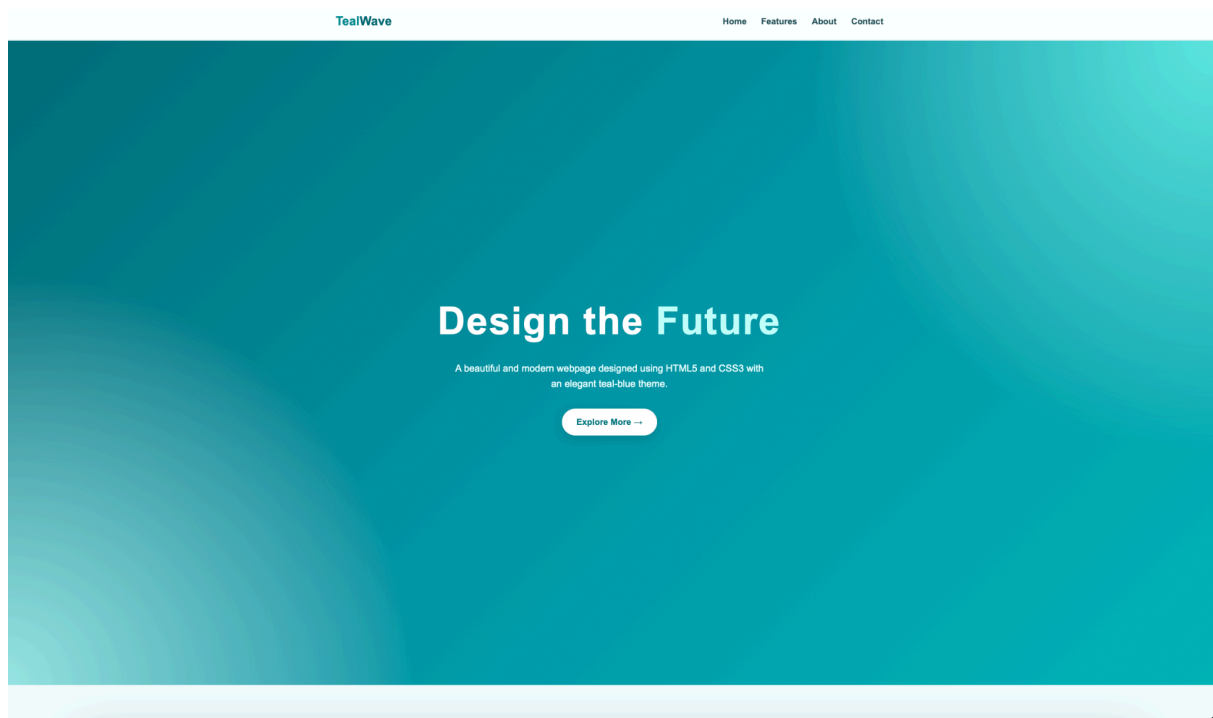
CSS Transitions	Provides smooth visual changes during hover interactions.
CSS Hover Effects	Changes the appearance and position of elements when the user hovers over them.
Media Queries	Makes the webpage responsive for tablets and mobile devices.
<footer>	Defines the footer section containing copyright information.

8. TEST CASES AND EXECUTION DATA

Test Case	Test Scenario	Input / Action	Expected Result	Actual Result	Status
TC-01	Open webpage	Open <code>index.html</code> in browser	TealWave webpage should load successfully	Webpage loaded correctly	Pass
TC-02	Check navigation bar	Click Home, Features, About, and Contact	Page should navigate to the respective sections	Navigated displayed correctly	Pass
TC-03	Check HTML5 Canvas	Open Home section	Heading, description, and Explore More button should be displayed	Hero section displayed correctly	Pass
TC-04	Check Features section	Open Features section	Three feature cards should be displayed	All three cards displayed correctly	Pass
TC-05	Check card hover effect	Move cursor over a feature card	Card should move upward and show a shadow effect	Hover effect worked correctly	Pass

TC-06	Check Explore More button	Click Explore More →	Page should move to Features section	Navigated to Features section	Pass
TC-07	Check About section	Open About section	About content and HTML5 + CSS3 information should be displayed	About section displayed correctly	Pass
TC-08	Check Contact button	Click Contact Us	Email link should open the mail application	Mail link responded correctly	Pass
TC-09	Check responsive layout	Resize browser window below 800px	Cards and About section should change to a single-column layout	Layout adjusted correctly	Pass
TC-10	Check mobile layout	Resize browser below 500px	Navigation should wrap and spacing should adjust	Mobile layout displayed correctly	Pass

9. OUTPUT



10. RESULT

Thus, a **modern responsive webpage named TealWave** was successfully designed using **HTML5 and CSS3**. Navigation, hero section, feature cards, About section, Contact section, gradients, hover effects, transitions, and responsive layouts were implemented successfully and verified using multiple test cases.